Day 2 Activity: Hospital Staffing & Medical Protocols (Intro Stats)

Name: _____ Date: _____

Scenario: You are the scheduling coordinator for a busy hospital. Your job involves organizing staff schedules, patient care pathways, and medication administration procedures. Use counting principles to answer each question.

Part 1: Fundamental Counting Principle (FCP)

Show your setup as a product (e.g., choices × choices × choices) and give the final number.

1) Night Shift Staffing Select 1 doctor from 4, 1 registered nurse from 6, and 1 medical assistant from 3 for the Emergency Department night shift. How many different staffing combinations are possible? Work:

2) Chest Pain Care Pathway A patient can choose 1 of 3 cardiologists, then 1 of 2 stress tests, then 1 of 4 cardiac nurses. How many possible care pathways are there? Work:

3) Medication Administration Protocol A nurse verifies identity using 1 of 2 methods, checks the medication using 1 of 3 verification systems, and documents using 1 of 2 electronic systems. How many protocol sequences exist? Work:

Part 2: Simplifying Factorial Expressions (Numbers Only)

These mirror the calculations inside permutation and combination formulas. Simplify each exactly (no calculator for final form unless specified by your instructor).

4) $8!/5! =$

5) $10!/(7! \cdot 3!) =$

6) $12!/9! =$

7) $9!/(6! \cdot 3!) =$

8) $15!/(12! \cdot 3!) =$

Part 3: Permutations (Order Matters)

Identify the correct model and compute.

9) Medication Order A patient must receive 5 different medications in a specific sequence to avoid interactions. In how many different orders can the 5 medications be given? Model/Work:

10) Surgical Roles From a surgical team of 8 professionals, assign 3 specific roles: Lead Surgeon, First Assistant, and Anesthesiologist. In how many ways can these roles be filled? Model/Work:

Part 4: Combinations (Order Does Not Matter)

Identify the correct model and compute.

11) Patient Safety Committee Select 4 nurses from a group of 10 volunteers to form a committee. How many different committees are possible? Model/Work:

12) Clinical Trial Enrollment Choose 5 patients from 12 volunteers to participate in a clinical trial. How many different groups of 5 can be selected? Model/Work:

Optional Reflection (If time remains) • When should you use FCP instead of permutations or combinations? • How do you recognize when order matters in a real medical context?